

AI-to-Lab Orchestrator

The operating layer between an AI's next-experiment suggestion and the evidence that it actually ran.

2

use cases, both defensible

65

tests, including the governance and evidence layers

30 x 30

seed benchmark, reproducible from one script

A self-driving-lab orchestration prototype for materials discovery · not a physics simulator

Predictions outpace validation

Models propose candidates far faster than labs can evaluate them. The bottleneck is not model accuracy — it is that a prediction does not automatically become an experiment, an experiment does not automatically become trustworthy data, and trustworthy data does not automatically become a decision.

**What should we try next?**

and by whose definition of a good result

**Is it safe to run?**

individually legal values can be jointly unsafe

**Did it actually run?**

or did it fail, and was the failure recorded

**Is the data good enough?**

not every completed run deserves to be believed

**What did we learn?**

which hypothesis did this settle, if any

Everything in that gap is operational.

Scheduling. Validation. Safety review. Provenance. Data quality.
Deciding what is even eligible to feed back into a model.

None of it is a modelling problem. All of it decides how fast the loop actually turns — and the speed of that loop is the ceiling on how fast an AI-for-science programme can iterate.

That gap is what I built.

Five layers, each answering one question

Data contracts

If a device changes its output shape, do we find out?

Partial**Orchestration**

How does an intent become an auditable sequence of steps?

Done**Experiment data**

What is the system's memory? Do failures count as data?

Partial**Decision**

What next, and by whose definition of “good”?

Done**Governance**

Who approved this, and which data may train a model?

Done

Separated so each can be reasoned about — and swapped — on its own. The optimizer sits behind a two-method interface; the dashboard renders state and owns no workflow logic; orchestration and decisions run headless without importing Streamlit.

Scientific honesty over fake physics

Where physical fidelity matters, use published data. Where systems benchmarking matters, use a transparent surrogate that is honest about being one. The two use cases were chosen to sit on opposite sides of that line.

TRANSPARENT SURROGATE

CdTe thin-film process

- A benchmark environment for orchestration and optimization
- Deliberately hard: observation noise, a narrow non-smooth treatment window, an over-treatment cliff, parameter interactions, outright failures
- Literature-inspired in qualitative behaviour only

What it is not

- A physically accurate CdTe simulator
- Any claim about real device efficiency

PUBLISHED DATA

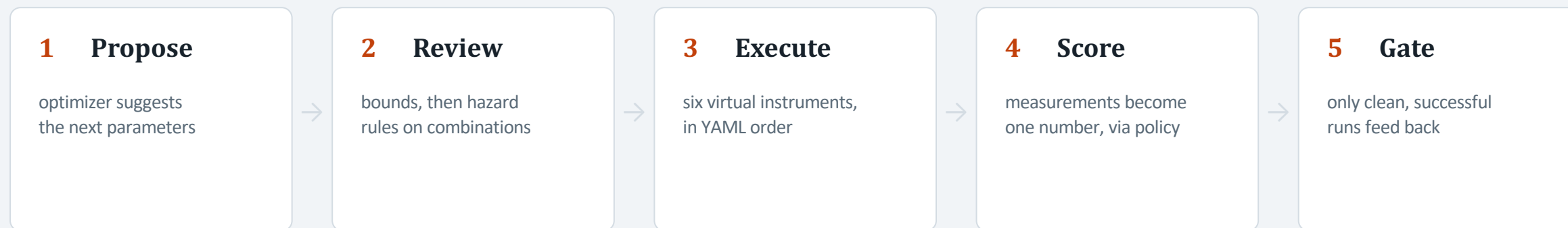
Hydride superconductors

- 22 candidates transcribed verbatim from Sanna et al., Communications Physics (2026)
- Triage, decision policy and validation planning built on top of published values
- Provenance enforced as a file boundary, with a test to hold it

What it is not

- Any simulation of DFPT, electron-phonon coupling or T_c
- Any score the paper did not publish, presented as though it had

The closed loop, and what it refuses to do



blocked ≠ failed

A failure means a sample was consumed and produced nothing. A block means the gate refused to consume one at all. They cost a real lab completely different amounts, so they are separate statuses and the benchmark reports them separately.

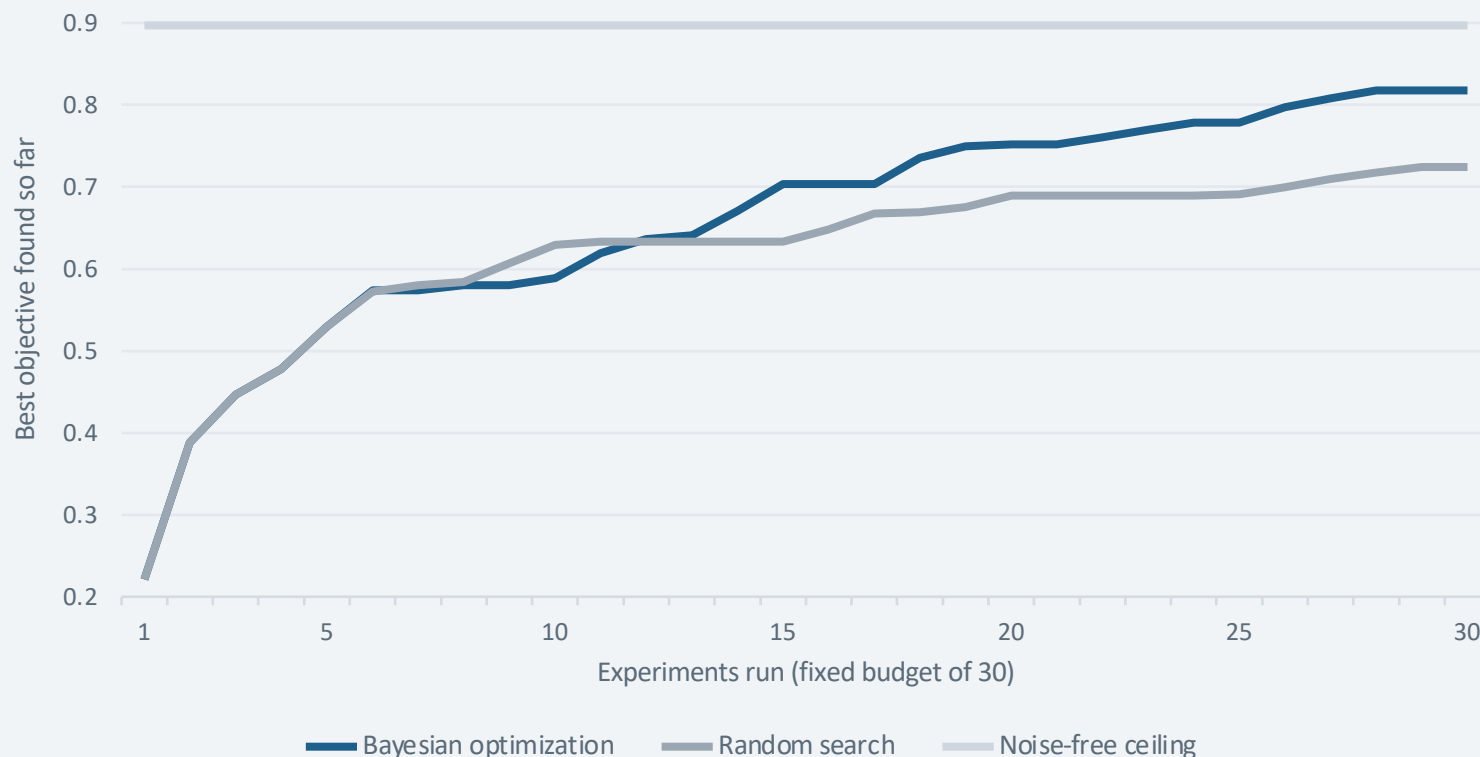
A blocked experiment runs zero devices.

The guardrail is set inside the uncertainty band

`blocks at 440 °C / 40 min`
`degrades at 450 °C / 45 min`

A real lab does not know exactly where the cliff is, so the rule fires before the sample is destroyed. This costs reachable search space — which is why blocked is reported as its own rate. Governance is not free, and the system shows its price.

Bayesian optimization vs random, 30 seeds



0.818

Bayesian optimization, median best
IQR [0.723, 0.855]

0.724

Random search, median best
IQR [0.663, 0.806]

20 / 30

seeds where BO won, median gap +0.093
Paired comparison is valid because both
methods meet the same lab noise.

Three things I say before I am asked

1

The bands overlap

BO wins in the median and on two seeds in three, but it does not dominate. On a five-dimensional landscape with a 30-experiment budget, clean separation would be evidence the benchmark was too easy — not evidence the optimizer is good.

2

BO is behind early

It runs an initial design first, so it trails random search for roughly the first ten experiments. The advantage is sample efficiency later in the budget, not from experiment one. A chart cropped at 10 would tell the opposite story.

3

Runs exceed the ceiling

The dashed line is the noise-free maximum, but the reported score is a noisy observation — so best-found is an optimistically biased estimator. Taking a max over noisy draws captures luck. Any lab ranking candidates on a single best-observed inherits this, which is why replication is a governance concern.

A portfolio that only shows what worked is asking to be caught. Putting the weaknesses in the product surface itself — they are printed in the dashboard and the README — is cheaper than being asked.

The weights are a policy, not a constant

How much is phase purity worth relative to efficiency? That is a research-strategy question, not a property of an XRD machine. So the weights live in versioned policy files, every experiment records which policy scored it, and the same mechanism drives hydride triage.

Policy	Median best	Treatment time	Treatment temp	Dopant
Performance first	0.773	35.9 min	387.4 °C	1.2 %
Balanced device quality	0.818	40.0 min	387.6 °C	1.3 %
Manufacturability first	0.926	27.7 min	388.6 °C	0.9 %

Median best is not comparable across these rows. Each policy defines a different objective, so a higher number does not mean a better process. The comparable quantity is the operating point: manufacturability-first converges on a treatment twelve minutes shorter and lighter doping — a visibly more conservative process. Treatment temperature lands near 388 °C under every policy, because that window dominates the landscape. Policy influences the parameters the objective leaves room to argue about.

Published data, and judgment kept visible

The paper publishes λ , ω_{log} and an Allen–Dynes Tc. It publishes no score, no confidence and no feasibility. Ranking needs those — so they are computed from documented rules and kept in separate files. Provenance is a file boundary, not a convention, and a test fails if it erodes.

PUBLISHED`datasets/hydrides/`

Transcribed verbatim from Tables 1–2.
 $\lambda = 1.00 \cdot \omega_{\text{log}} = 341.59 \text{ K} \cdot \text{Tc} = 23.5 \text{ K}$

DERIVED`trriage/derive.py`

A documented transform of published values.
Tc confidence, from coupling regime and whether a refinement exists.

ANALYST`configs/trriage/`

Judgment encoded as a versioned rule set.
Synthesis feasibility, from elemental constraints. Labelled as judgment.

Only 1 of 22 candidates has a beyond-Allen–Dynes Tc.

For that one, refinement moved $23.5 \text{ K} \rightarrow 17 \text{ K}$, which the authors call “an uncommon deviation”. Every other ranking rests on a number the paper’s own authors caution against over-reading — so confidence is weighted separately from Tc, rather than folded invisibly into it.

The finding a Tc-ordered list cannot show

10 of 22

candidates contain technetium

It has no stable isotope and needs a licensed facility. That constraint is visible from the formula alone and invisible in any ranking sorted by predicted Tc. Feasibility combines as a minimum, not a mean — one disqualifying element is not offset by three convenient ones.

Ranks move with the policy

Candidate	Bal.	High-Tc	Feas.
LiZrH6Ru	1	1	1
Ta6MoH16	2	3	2
EuCdH6Ru	6	2	9
EuLuTcH6	13	5	14

EuCdH6Ru moves 2 → 9 between policies — the warning label on every other row.

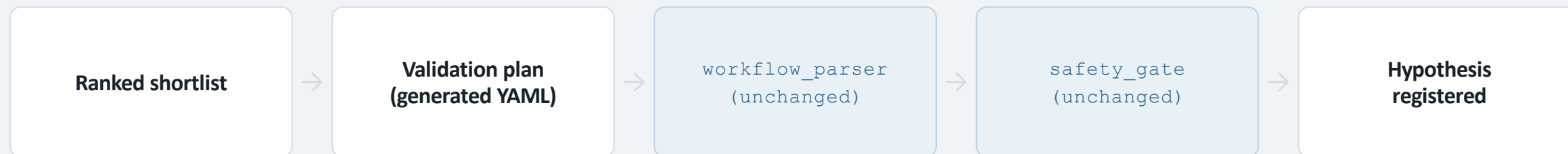
The consensus shortlist is the useful output, not rank 1.

LiZrH6Ru · Ta6MoH16 · TaNb3H8 — these sit in the top five under every policy considered. They are what a lab validates regardless of whose priorities win the argument, which is the question a programme lead actually has to answer.

A single ranking presented as the answer would manufacture authority the data does not carry. The sensitivity view and the consensus shortlist are the honest output.

One system, not two demos in one repository

Triage does not end at a ranked table. The selected candidate becomes a workflow that enters the same machinery a CdTe experiment does — same parser, same safety gate, no orchestrator changes.



The CdTe safety gate blocked a hydride hazard it had never seen.

750 °C with 150 bar H₂ → blocked: hydrogen_pressure_at_temperature

Both values are individually within bounds. The generated workflow declares its own hazard rules and the existing gate enforces them correctly on first sight. A test asserts this — if it ever fails, the project has gone back to being two demos in one repository.

Plans stop at awaiting_device_implementation — no hydride synthesis is simulated, because there is no honest way to. That is the forward path. What comes back across the same seam is the next slide.

A measurement changes the ranking that proposed it

Evidence is received, never manufactured. There are no hydride devices, so records enter from outside — an instrument, a collaborator, a paper. A record with no named recorder or stated source is rejected at load: a measurement nobody will sign for cannot overturn a prediction.

A null result is meaningless without its floor

Ta6MoH16: no transition seen — but the rig reached only 8 K and the prediction is 7.3 K. The run could not have observed what it was testing, so it refutes nothing. Marked inconclusive, excluded from the calibration, and its hypothesis stays open.

Evidence about one compound calibrates the method

21 of 22 candidates rest on Allen–Dynes alone. On 2 conclusive measurement(s), Allen–Dynes runs 28% high (median observed/predicted = 0.72). Confidence in every un-measured Allen–Dynes prediction is reduced by 14%. A calibration lowers confidence and never rewrites a prediction — a "corrected" Tc would invent a number nobody computed.

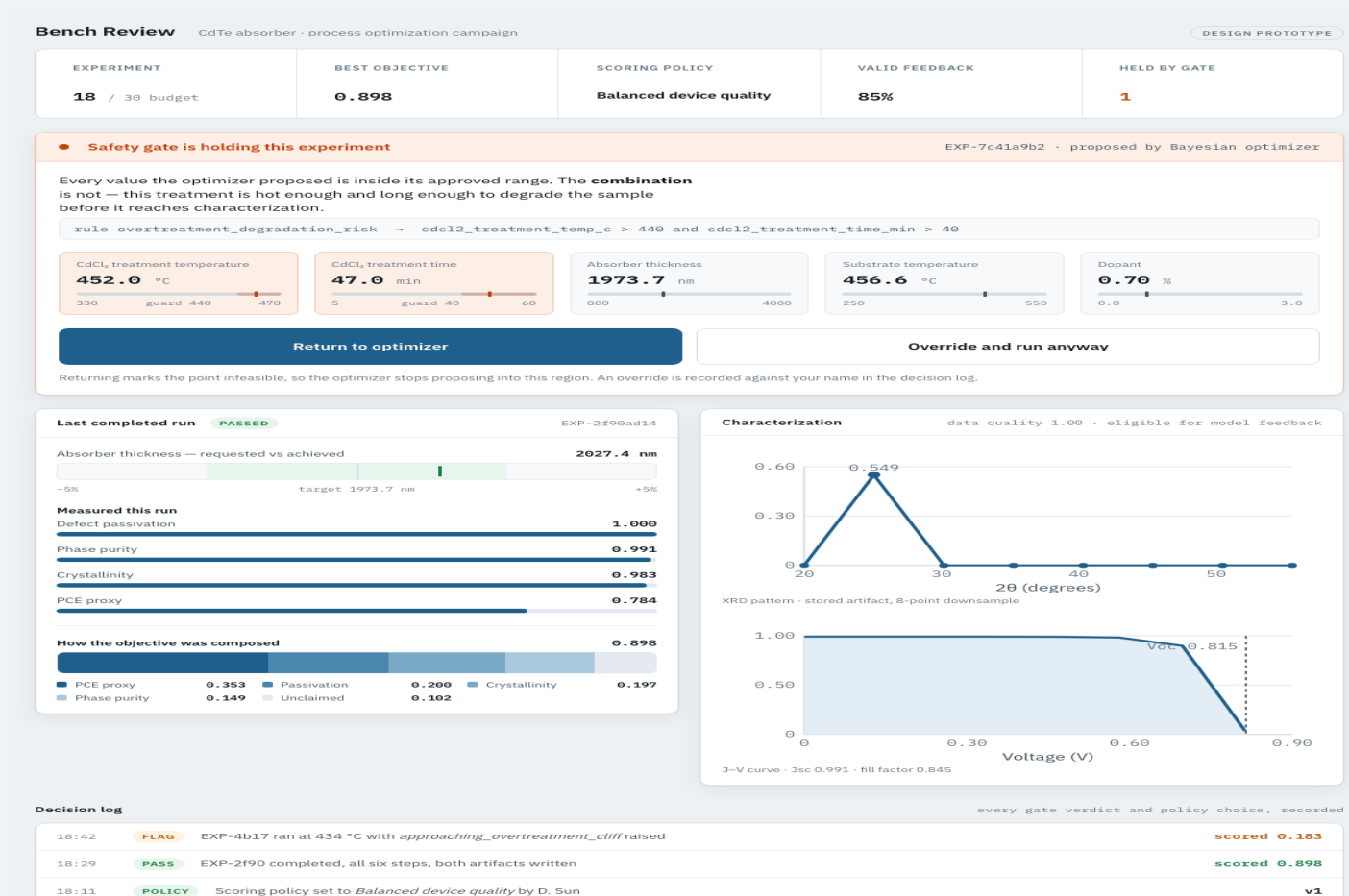
The cohort re-ranks

Candidate	was	now	Tc source
LiZrH6Ru	1	1	measured
TaNb3H8	3	2 (+1)	measured
EuCdH6Ru	6	3 (+3)	Allen–Dynes
Ta6MoH16	2	4 (–2)	Allen–Dynes
Ta3NbH8	4	5 (–1)	Allen–Dynes
Ta6NbH16	5	6 (–1)	Allen–Dynes

A measured Tc supersedes every calculation, and the hypothesis moves from proposed to supported or contradicted.

Measuring the leader low moved candidates nobody tested. Tc is normalized against the best in the cohort, so when the leader falls every untested compound becomes relatively more attractive — partly offset by the confidence penalty the same evidence applied to the method. A ranking is a statement about a set, not about a compound.

Where a person actually meets the loop



The approval queue is the hero

The held experiment is the only thing on the loop that needs a human. Everything else — proposing, executing, logging — does not. Putting approval anywhere but first would mean missing why this screen exists.

Every value is real output

The held parameters, the triggered rule, the measurements, the objective breakdown and both artefacts are genuine system output, not illustration.

Sized for gloved hands

60px touch targets, no text entry, high contrast. A tablet at the bench, not a dashboard at a desk.

A high-fidelity prototype is a product manager's deliverable. Building it in React would have demonstrated wanting to be a front-end engineer instead.

Five defects I found in my own build

These are real, not hypotheticals, and they are documented in the repository. Four of the five were silent — the system looked correct while being wrong.

01 The safety gate protected nothing

Its hazard region was a strict subset of the true failure region, so every experiment it flagged was already doomed. It prevented zero damage while appearing to work in every demo.

Governance fails identically to how it succeeds. It needs a test, not a demo.

02 The benchmark numbers were wrong

The README claimed a BO median of 0.84. The measured value was 0.806, and no script in the repository produced either number.

One script is now the sole source of every published figure.

03 One RNG served lab and optimizer

BO draws ~512 candidates per iteration, random search draws 5 — so the two methods faced different measurement noise. The per-seed comparison was meaningless.

Separate streams, and common random numbers across methods.

04 BO got trapped in failure regions

Failed experiments yield no objective, so a naive surrogate never learns to avoid them and keeps re-proposing into the dead zone, burning budget.

A feasibility penalty and an escape hatch — a real constrained-BO problem.

05 Rounding moved the median by 0.006

Rounding each weighted term to six decimals — a $1e-4$ perturbation — changed which point the GP held as incumbent, changed the argmax of Expected Improvement, and redirected the whole search.

Closed loops are chaotically sensitive to their own numerics. Round for display, never before a decision.

What I deliberately did not build

Scope discipline is part of the argument, so the omissions are explicit and reasoned rather than quietly absent.

- | | | |
|---|--|---|
| ✗ | A physically accurate CdTe simulator | Would need real process data. A fake physics model is worse than an honest surrogate. |
| ✗ | Simulated DFPT, electron-phonon or Tc | Same reason. The hydride module uses published values and builds decisions on top. |
| ✗ | A graphene tactile-sensor use case | No grounded response model available. Two defensible use cases beat three thin ones. |
| ✗ | Hydride synthesis devices | No honest way to simulate a hydrogenation anneal. Plans stop where a real lab's seam is. |
| ✗ | A real async task queue | The state machine is what matters. The infrastructure adds operational surface, not argument. |
| ✗ | A “correct” hydride ranking | The paper publishes no scores. One ranking as the answer would manufacture authority. |

I am not switching careers. I am moving a proven delivery capability into a new domain.

Life-science consulting

A lab's bottleneck is usually its process and feedback loops, not a single technique. Optimizing R&D workflows is what I did.

0→1 founder

Built and shipped end to end under ambiguity and resource constraints, owning the outcome. This orchestrator is the same instinct in a new domain.

Platform growth at scale

Coordinated engineering, product and data teams, and translated technical work into measurable impact — the research-to-impact translation this work centres on.

What this prototype adds is the domain-technical layer: I understand the closed loop, the data, and the governance well enough to run programmes in it.